

# Ultra-Small, Low-Power, 16-Bit Analog-to-Digital Converter with Internal Reference

Check for Samples: [ADS1113](#) [ADS1114](#) [ADS1115](#)

## FEATURES

- **ULTRA-SMALL QFN PACKAGE:**  
2mm × 1,5mm × 0,4mm
- **WIDE SUPPLY RANGE:** 2.0V to 5.5V
- **LOW CURRENT CONSUMPTION:**  
Continuous Mode: Only 150µA  
Single-Shot Mode: Auto Shut-Down
- **PROGRAMMABLE DATA RATE:**  
8SPS to 860SPS
- **INTERNAL LOW-DRIFT VOLTAGE REFERENCE**
- **INTERNAL OSCILLATOR**
- **INTERNAL PGA**
- **I<sup>2</sup>C™ INTERFACE:** Pin-Selectable Addresses
- **FOUR SINGLE-ENDED OR TWO DIFFERENTIAL INPUTS (ADS1115)**
- **PROGRAMMABLE COMPARATOR (ADS1114 and ADS1115)**
- **OPERATING TEMPERATURE:** –40°C to +140°C

## APPLICATIONS

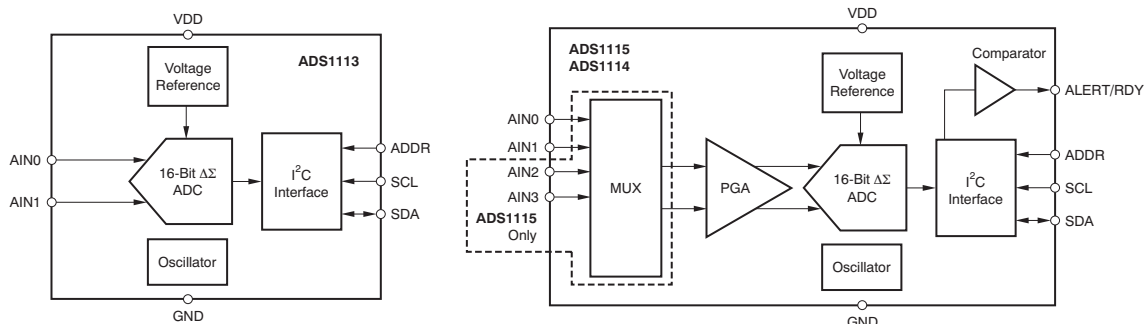
- **PORTABLE INSTRUMENTATION**
- **CONSUMER GOODS**
- **BATTERY MONITORING**
- **TEMPERATURE MEASUREMENT**
- **FACTORY AUTOMATION AND PROCESS CONTROLS**

## DESCRIPTION

The ADS1113, ADS1114, and ADS1115 are precision analog-to-digital converters (ADCs) with 16 bits of resolution offered in an ultra-small, leadless QFN-10 package or an MSOP-10 package. The ADS1113/4/5 are designed with precision, power, and ease of implementation in mind. The ADS1113/4/5 feature an onboard reference and oscillator. Data are transferred via an I<sup>2</sup>C-compatible serial interface; four I<sup>2</sup>C slave addresses can be selected. The ADS1113/4/5 operate from a single power supply ranging from 2.0V to 5.5V.

The ADS1113/4/5 can perform conversions at rates up to 860 samples per second (SPS). An onboard PGA is available on the ADS1114 and ADS1115 that offers input ranges from the supply to as low as ±256mV, allowing both large and small signals to be measured with high resolution. The ADS1115 also features an input multiplexer (MUX) that provides two differential or four single-ended inputs.

The ADS1113/4/5 operate either in continuous conversion mode or a single-shot mode that automatically powers down after a conversion and greatly reduces current consumption during idle periods. The ADS1113/4/5 are specified from –40°C to +125°C.



Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

I<sup>2</sup>C is a trademark of NXP Semiconductors.

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This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

## ORDERING INFORMATION

For the most current package and ordering information, see the Package Option Addendum at the end of this document, or see the TI web site at [www.ti.com](http://www.ti.com).

## ABSOLUTE MAXIMUM RATINGS <sup>(1)</sup>

|                                          | ADS1113, ADS1114, ADS1115 | UNIT |
|------------------------------------------|---------------------------|------|
| VDD to GND                               | –0.3 to +5.5              | V    |
| Analog input current                     | 100, momentary            | mA   |
| Analog input current                     | 10, continuous            | mA   |
| Analog input voltage to GND              | –0.3 to VDD + 0.3         | V    |
| SDA, SCL, ADDR, ALERT/RDY voltage to GND | –0.5 to +5.5              | V    |
| Maximum junction temperature             | +150                      | °C   |
| Operating temperature range              | –40 to +140               | °C   |
| Storage temperature range                | –60 to +150               | °C   |

(1) Stresses above those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. Exposure to absolute maximum conditions for extended periods may affect device reliability.

## PRODUCT FAMILY

| DEVICE  | PACKAGE<br>DESIGNATOR<br>MSOP/QFN | RESOLUTION<br>(Bits) | MAXIMUM SAMPLE<br>RATE (SPS) | COMPARATOR | PGA | INPUT CHANNELS<br>(Differential/<br>Single-Ended) |
|---------|-----------------------------------|----------------------|------------------------------|------------|-----|---------------------------------------------------|
| ADS1113 | BROI/N6J                          | 16                   | 860                          | No         | No  | 1/1                                               |
| ADS1114 | BRNI/N5J                          | 16                   | 860                          | Yes        | Yes | 1/1                                               |
| ADS1115 | BOGI/N4J                          | 16                   | 860                          | Yes        | Yes | 2/4                                               |
| ADS1013 | BRMI/N9J                          | 12                   | 3300                         | No         | No  | 1/1                                               |
| ADS1014 | BRQI/N8J                          | 12                   | 3300                         | Yes        | Yes | 1/1                                               |
| ADS1015 | BRPI/N7J                          | 12                   | 3300                         | Yes        | Yes | 2/4                                               |